

EXPERIENCES

THE HOME DEPOT

Jan 2024 – Present

Sr. Data Analyst — Delivery Analytics Team

Atlanta, GA

- Developed an ML Model achieving **85% accuracy** in detecting on-time delivery failures; collaborated with MLOps to productionize and integrate it into sourcing endpoints for a pilot, and conducted **A/B tests** before full rollout.
- Collaborated with data Scientists and software Engineers to develop a **SmartFulfill agent** using offline RL, optimizing fulfillment decisions and **reducing on-time delivery failures by 4.5%, retaining \$10M in customer spending**.
- Leveraged LLM to pinpoint the root cause of On time and Complete delivery failure type and enhanced accuracy of previous root cause analysis by 9% through thematic analysis of store associates' and drivers' comments.
- Collaborated with GenAI team to create a Gemini Agent chatbot to answer adhoc requests from logistics operation team and run BigQuery on the backend, using LangChain as the framework.

Data Analyst — Delivery Analytics Team

- Developed a SQL workbook to calculate recommended pallet counts, validated results through an API endpoint, and constructed a **pallet compliance dashboard** for stakeholders, potentially generating **annual savings of \$2 million**.
- Optimized the truck type compliance dashboard, reducing load time from 226 to 2.56 seconds (**100X improvement**).
- Performed **statistical analysis on 10 million orders** for R12M to determine key factors influencing on-time delivery.
- Developed Dashboard to track the Ship from best location initiative introduced by leadership and found 500K more order opportunity across different ship type to improve speed to customer
- Developed a Python-based **"Control Tower"** to track vehicle rule changes and enhance Truck Type Model accuracy.
- Collaborated with Online Delivery Analytics team to devise an optimization model to pinpoint the optimal carrier for each vendor, maximizing discounts from the primary carriers and **potentially yielding over \$30 million in savings**.
- Consistently delivered on multiple ad-hoc requests, including effectiveness analysis of new sprinter vans, rural area box additions, and extraction of pallet data for MDS.

VOLVO GROUP

Jan 2023 – Dec 2023

Data Analyst — Logistics Purchasing Team

Greensboro, NC

- Optimized the load time of the logistics supplier's performance Power BI dashboard by 40% using DAX in Power BI.
- Engineered **10+ Power Automate flows, 5+ Power Apps, and 5+ Power BI dashboards** for process visualization, saving over 300 hours annually.
- Constructed a Power Automate flow for invoice data analysis, identifying \$200K in savings on a \$1.8M dataset.
- Utilized SQL for large-scale data extraction and applied time-series and machine learning to enhance weekly inbound shipment predictions, improving model accuracy by 8% within three months.

SYSTEMS ANALYTICS LAB, WEST VIRGINIA UNIVERSITY

Aug 2021 – Dec 2022

Data Science Research Assistant

Morgantown, WV

- Increased prediction accuracy by 10% by analyzing tunnel heating issues using Python and time-series ML algorithms.
- Developed a hybrid deep neural network for a track association in marine surveillance, achieving 85% accuracy.
- Performed data analysis on patient datasets and clustered similar patient groups using PCA and K-Means algorithms.

SKILLS

Technical Skills : SQL (Google Bigquery), Tableau, Python (Pandas, Scikit-learn, NumPy, Matplotlib, TensorFlow, Pytorch) , CPLEX, GCP, Analytics Engine, Vertex AI, Power BI, Power Apps, MS Excel (Pivot Table, Power Query, VBA, Macros), PowerPoint, MS Project, Git, SAP ERP and Power Automate.

Certifications : Microsoft Certified Power BI Data Analyst (PL-300), Certified Six Sigma Green Belt (IISE)

Awards : **Bronze Medalist** for The Home Depot Data Science and Analytics **Hackathon(2024)**, Finalist of Data Challenge for Quality Control and Reliability Engineering (IISE)

EDUCATION

Georgia Institute of Technology

2025

MS in Computer Science (specialization in Machine Learning)